

The Receptor-Like Cytoplasmic Kinase (*OsRLCK*) Gene Family in Rice: Organization, Phylogenetic Relationship, and Expression during Development and Stress

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ABSTRACT Receptor-like cytoplasmic kinases (RLCKs) in plants belong to the super family of receptor-like kinases (RLKs). These proteins show homology to RLKs in kinase domain but lack the transmembrane domain. Some of the functionally characterized RLCKs from plants have been shown to play roles in development and stress responses. Previously, 149 and 187 RLCK encoding genes were identified from *Arabidopsis* and rice, respectively. By using HMM-based domain structure and phylogenetic relationships, we have identified 379 *OsRLCKs* from rice. *OsRLCKs* are distributed on all 12 chromosomes of rice and some members are located on duplicated chromosomal segments. Several *OsRLCKs* probably also undergo alternative splicing, some having evidence only in the form of gene models. To understand their possible functions, expression patterns during landmark stages of vegetative and reproductive development as well as abiotic and biotic stress using microarray and MPSS-based data were analyzed. Real-time PCR-based expression profiling for a selected few genes confirmed the outcome of microarray analysis. Differential expression patterns observed for majority of *OsRLCKs* during development and stress suggest their involvement in diverse functions in rice. Majority of the stress-responsive *OsRLCKs* were also found to be localized within mapped regions of abiotic stress QTLs. Outcome of this study would help in selecting organ/development stage specific *OsRLCK* genes/targets for functional validation studies.

Key words: abiotic stress; biotic stress; genome-wide analysis; kinase; rice; RLCK.

INTRODUCTION

The receptor-like protein kinases (RLKs) form one of the largest and most diverse superfamilies of plant proteins, with ~600 reported members in *Arabidopsis* and ~1131 members in the rice genome (Shiu et al., 2004). Recently, Dardick et al. (2007) have generated a rice kinase database comprising 1429 proteins. A typical RLK is a transmembrane protein with an extracellular receptor domain for signal perception and an intracellular kinase domain for signal transduction (Shiu and Bleecker, 2001a). Plant RLKs show homology in the kinase domain to *Drosophila melanogaster* Pelle (involved in dorsal/ventral axis determination) and mammalian interleukin receptor-associated kinases involved in innate immunity (Shiu and Bleecker, 2001a). Plant RLKs characterized to date have been implicated in diverse biological processes, including development, self-incompatibility, response to pathogens, as well as multiple environmental stresses. A significant difference between plant and animal RLKs can be

seen in their number, with animals containing one to six members, whereas, in plants, their number is 100–200-fold higher (Shiu and Bleecker, 2003). Previous reports have confirmed that lineage-specific expansion (LSE) is particularly evident in plants, especially in relation to certain gene families such as those involved in pathogen and stress response, transcription regulation, ubiquitin-mediated protein degradation system, protein modification, signal transduction, chemoreception, and small molecule metabolism (non-polymeric, non-protein biomolecules) (Lespinet et al.,

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2002; Rice Annotation Project, 2007). The *RLK* gene superfamily in *Arabidopsis* and rice has been reported to include 147 (~25%) and 187 (~17%) members, respectively, which code for proteins lacking the transmembrane domain found in the typical RLKs but are nevertheless grouped in this superfamily due to their apparent homology. These have been named Receptor-Like Cytoplasmic Kinases (RLCKs) (Shiu et al., 2004). Some well characterized RLCKs include tomato *Pto* (disease resistance), *Arabidopsis CDG1* (growth and differentiation), *Arabidopsis CRCK1* (abiotic stress response), *Arabidopsis PBS1* (disease resistance), and wheatgrass *Esi47*, stress-responsive gene involved in hormone signaling (Tang et al., 1996; Shen et al., 2001; Swiderski and Innes, 2001; Muto et al., 2004; Yang et al., 2004).

Another functionally important class of related proteins, called RLPs (receptor-like proteins), is not categorized as RLKs because it lacks the kinase domain but has the extracellular receptor domain (Jones et al., 1994; Kayes and Clark, 1998; Wang et al., 1998; Jeong et al., 1999; Nadeau and Sack, 2002; Shiu and Bleecker, 2003; Tamura et al., 2003; Verica et al., 2003; Ron and Avni, 2004; Fritz-Laylin et al., 2005). Thus, RLCKs as well as RLPs seem to be the variants of typical RLKs, with the former being similar to the kinase domain while the latter to the receptor domain. Interestingly, some studies have shown that RLKs, RLPs, and RLCKs could function in concert in similar signaling pathways (Zhang et al., 2005). While an in-depth analysis has been carried out for *RLP* and *RLK* genes in rice (Fritz-Laylin et al., 2005; Morillo and Tax, 2006), such an analysis has not been carried out/attempted for RLCKs and, except for a few, the biological relevance of the majority of RLCKs, especially vis-à-vis a typical RLK, still remains unresolved. As an initial step to understand the role of the RLCKs as a gene family, an in-silico analysis was performed using the finished level rice genome sequence (International Rice Genome Sequencing Project, 2005) to establish their number, chromosomal localization, duplication pattern, domain composition, as well as phylogenetic relatedness. Further, microarray, MPSS, and QTL data analysis were performed to study the expression profile and possible function of *OsRLCKs* in development and stress.

RESULTS

Identification of *OsRLCKs*

In order to establish the number of RLCKs in the rice genome, kinase domains of RLK subfamilies representative of rice and *Arabidopsis* and their homologs in diverse organisms (see Methods) were used for building an HMM profile. An additional HMM profile was built using *indica* rice (187) and *Arabidopsis* (147) RLCKs (Shiu et al., 2004). These were used for searching rice chromosomes pseudomolecules version 5 of TIGR. In order to identify the RLK superclade, the representative kinase domain sequences (Shiu et al., 2004) were aligned with HMM outcome and their interrela-

tionship viewed using NJ plot. A total of 375 receptor-like cytoplasmic kinases along with RLKs formed a superclade with significant bootstrap support (>60%). Four additional RLCKs were identified by performing a similar search in the KOME database. Thus, in total, 379 RLCKs were identified in the rice genome.

To identify the extent of variability between *japonica* and *indica* rice for RLCKs, 379 *OsRLCKs* were used for BLAST search against the 187 RLCKs reported from the *indica* genome by Shiu et al. (2004). Only 100 *indica* RLCKs were found to have counterparts in the *japonica* *OsRLCK* at ≥80% coverage and ≥90% sequence identity levels. The *indica* RLCKs reported by Shiu et al. (2004) are derived from the rice genome published by Yu et al. (2002). Therefore, to account for the large difference in the number of RLCKs identified for *indica* and *japonica* rice genomes, the improved version of the *indica* rice genome (Yu et al., 2005) was also searched for homologs of *japonica* RLCKs. A total of 300 *indica* sequences were found to show a match to the *japonica* RLCK family, bringing the count closer to that in *japonica* rice.

Organization of the *OsRLCK* Gene Family

Analysis of the intron-exon organization of the *OsRLCKs* showed that the number of introns per gene varied from 0 to 26, with more than 60% containing at least five intervening sequences. The evidence of expression existed for more than 60% of the family members in terms of support from either an EST or a full-length cDNA (Table 1) and 88% have expression evidence from microarray and MPSS. Study of the domain organizations of the *OsRLCKs* revealed that more than 70% (285 *OsRLCKs*) had only a kinase domain while the rest (94 *OsRLCKs*) had additional domains similar to the extracellular receptor domains present in typical RLKs, such as LRR, lectin/EGF, DUF, U BOX, and WD40, among others. Representative domain organization is summarized in Figure 1 and Table 1.

The rice *OsRLCKs* were also mapped to TIGR pseudomolecules (version 5; chromosomes 1–12) based on coordinates of TIGR loci (<http://rice.plantbiology.msu.edu/pseudomolecules/info.shtml>) (Figure 2). The *OsRLCK* family was distributed on all rice chromosomes, with the maximum number (57) present on the largest chromosome 1 and least 15 on chromosome 8. A total of 59 genes were identified in duplicated segments; out of these, 46 were paired RLCKs and, for 13 genes, duplicate partners were not RLCKs and, therefore, not represented as members of the *OsRLCK* family (Figure 2 and Supplemental Table 1). Two *OsRLCK* genes were considered tandemly duplicated if they were separated by fewer than five intervening genes and shared ≥40% sequence homology at protein level. A total of 49 *OsRLCKs* were thus found to be tandemly duplicated, falling into 21 groups, out of which 16 groups contained two members while others had more than two members. Maximum numbers of tandemly duplicated *OsRLCKs* were located on chromosome 1 and 11 (Supplemental

Table 1. Features of OsRLCKs present in the rice genome.

Gene ID	Locus ID*	Size (bp)	Introns (No.)	EST/ cDNA ⁺	Size (aa)	Domain [#]
<i>OsRLCK1</i>	LOC_Os01g01110.1	894	1		298	a
<i>OsRLCK2</i>	LOC_Os01g02290.1	1854	2	C	618	a
<i>OsRLCK3</i>	LOC_Os01g02300.1	1980	2	C	660	a
<i>OsRLCK4</i>	LOC_Os01g02320.1	1512	2	C	504	a
<i>OsRLCK5</i>	LOC_Os01g02390.1	1980	2	C	660	a
<i>OsRLCK6</i>	LOC_Os01g02400.1	1869	2	E	623	a
<i>OsRLCK7</i>	LOC_Os01g02410.1	2136	3	E	712	a
<i>OsRLCK8</i>	LOC_Os01g02430.1	1893	1	E	631	a
<i>OsRLCK9</i>	LOC_Os01g02460.1	915	0		305	a
<i>OsRLCK10</i>	LOC_Os01g02550.1	2202	6		734	a
<i>OsRLCK11</i>	LOC_Os01g02560.1	1866	2	C	622	a
<i>OsRLCK12</i>	LOC_Os01g02580.1	1941	2		647	a
<i>OsRLCK13</i>	LOC_Os01g02680.1	2133	2	C	711	a
<i>OsRLCK14</i>	LOC_Os01g02690.1	1926	2	E	642	a
<i>OsRLCK15</i>	LOC_Os01g02760.1	2139	4		713	a
<i>OsRLCK16</i>	LOC_Os01g02780.1	2172	2	C	724	a
<i>OsRLCK17</i>	LOC_Os01g02790.1	1464	3		488	a
<i>OsRLCK18</i>	LOC_Os01g02800.1	2058	2	E	686	a
<i>OsRLCK19</i>	LOC_Os01g03690.1	546	1	C	182	a
<i>OsRLCK20</i>	LOC_Os01g04480.1	1869	3	E	623	a
<i>OsRLCK21</i>	LOC_Os01g04490.1	1854	5	E	618	a
<i>OsRLCK22</i>	LOC_Os01g04520.1	1881	4		627	a
<i>OsRLCK23</i>	LOC_Os01g04550.1	870	1	E	290	a
<i>OsRLCK24</i>	LOC_Os01g05940.1	4233	7	E	1411	d
<i>OsRLCK25</i>	LOC_Os01g05960.1	3156	1	C	1052	d
<i>OsRLCK26</i>	LOC_Os01g10700.1	2253	2		751	f1
<i>OsRLCK27</i>	LOC_Os01g14510.1	1053	4	C	351	a
<i>OsRLCK28</i>	LOC_Os01g15470.1	1368	4	C	456	a
<i>OsRLCK29</i>	LOC_Os01g16230.1	1191	0		397	a
<i>OsRLCK30</i>	LOC_Os01g19160.1	936	1	C	312	a
<i>OsRLCK31</i>	LOC_Os01g20880.1	2622	4	E	874	a
<i>OsRLCK32</i>	LOC_Os01g20900.1	2517	5	E	839	a
<i>OsRLCK33</i>	LOC_Os01g21970.1	1116	6	C	372	a
<i>OsRLCK34</i>	LOC_Os01g23970.1	1278	4		426	a
<i>OsRLCK35</i>	LOC_Os01g26210.1	1575	3	E	525	f2
<i>OsRLCK36</i>	LOC_Os01g36500.1	2547	11	E	849	c
<i>OsRLCK37</i>	LOC_Os01g38840.1	933	3		311	a
<i>OsRLCK38</i>	LOC_Os01g39820.1	723	2		241	a
<i>OsRLCK39</i>	LOC_Os01g39970.1	2298	4	C	766	h
<i>OsRLCK40</i>	LOC_Os01g40590.1	1485	5	C	495	a
<i>OsRLCK41</i>	LOC_Os01g41730.1	738	4	C	246	a
<i>OsRLCK42</i>	LOC_Os01g41870.1	2250	8	E	750	a
<i>OsRLCK43</i>	LOC_Os01g48020.1	2337	0		779	f3
<i>OsRLCK44</i>	LOC_Os01g57540.1	537	2	C	179	a
<i>OsRLCK45</i>	LOC_Os01g57940.1	1404	4	C	468	a
<i>OsRLCK46</i>	LOC_Os01g59560.1	2682	8		894	a
<i>OsRLCK47</i>	LOC_Os01g60700.1	1629	3	C	543	a
<i>OsRLCK48</i>	LOC_Os01g63280.1	1362	5	E	454	a

Table 1. Continued

Gene ID	Locus ID*	Size (bp)	Introns (No.)	EST/ cDNA ⁺	Size (aa)	Domain [#]
<i>OsRLCK49</i>	LOC_Os01g65940.1	1584	7	E	528	a
<i>OsRLCK50</i>	LOC_Os01g66040.1	753	1		251	a
<i>OsRLCK51</i>	LOC_Os01g66630.1	1611	3		537	a
<i>OsRLCK52</i>	LOC_Os01g67340.1	1110	6	C	370	a
<i>OsRLCK53</i>	LOC_Os01g70410.1	1095	6	E	365	a
<i>OsRLCK54</i>	LOC_Os01g70970.1	1155	6	C	385	a
<i>OsRLCK55</i>	LOC_Os01g71000.1	1476	4	C	492	a
<i>OsRLCK56</i>	LOC_Os01g73370.1	471	0		157	a
<i>OsRLCK57</i>	LOC_Os01g74200.1	1227	5	C	409	a
<i>OsRLCK58</i>	LOC_Os02g01730.1	1221	1	C	407	a
<i>OsRLCK59</i>	LOC_Os02g01800.1	3549	2		1183	d
<i>OsRLCK60</i>	LOC_Os02g02600.1	1284	5	C	428	a
<i>OsRLCK61</i>	LOC_Os02g05820.1	2328	9	C	776	h
<i>OsRLCK62</i>	LOC_Os02g06160.1	1950	2		650	a
<i>OsRLCK63</i>	LOC_Os02g06930.1	1203	5	E	401	a
<i>OsRLCK64</i>	LOC_Os02g09359.1	1134	3	C	378	a
<i>OsRLCK65</i>	LOC_Os02g12030.1	3399	1		1133	d
<i>OsRLCK66</i>	LOC_Os02g12130.1	540	4		180	a
<i>OsRLCK67</i>	LOC_Os02g12340.1	543	2		181	a
<i>OsRLCK68</i>	LOC_Os02g12660.1	2379	7	C	793	a
<i>OsRLCK69</i>	LOC_Os02g12670.1	2610	10	E	870	g2
<i>OsRLCK70</i>	LOC_Os02g13460.1	1605	5		535	d
<i>OsRLCK71</i>	LOC_Os02g16790.1	570	1		190	a
<i>OsRLCK72</i>	LOC_Os02g30900.1	1581	4	C	527	a
<i>OsRLCK73</i>	LOC_Os02g34430.1	1521	6	C	507	a
<i>OsRLCK74</i>	LOC_Os02g35760.1	1092	6	E	364	a
<i>OsRLCK75</i>	LOC_Os02g41480.1	1158	0	C	386	a
<i>OsRLCK76</i>	LOC_Os02g42160.1	2046	2		682	a
<i>OsRLCK77</i>	LOC_Os02g42190.1	2514	2		838	f2
<i>OsRLCK78</i>	LOC_Os02g42620.1	1110	5	E	370	a
<i>OsRLCK79</i>	LOC_Os02g43290.1	1098	1	C	366	a
<i>OsRLCK80</i>	LOC_Os02g43430.1	1398	3	C	466	a
<i>OsRLCK81</i>	LOC_Os02g44610.1	3249	14	E	1083	g3
<i>OsRLCK82</i>	LOC_Os02g44920.1	1377	4		459	a
<i>OsRLCK83</i>	LOC_Os02g45750.1	1089	1	E	363	a
<i>OsRLCK84</i>	LOC_Os02g53030.1	663	3	C	221	a
<i>OsRLCK85</i>	LOC_Os02g53720.1	1179	3	C	393	a
<i>OsRLCK86</i>	LOC_Os02g53750.1	1350	3	E	450	a
<i>OsRLCK87</i>	LOC_Os02g54590.1	2319	7	C	773	h
<i>OsRLCK88</i>	LOC_Os02g57420.1	1284	3	C	428	a
<i>OsRLCK89</i>	LOC_Os02g57440.1	1275	0		425	a
<i>OsRLCK90</i>	LOC_Os02g57560.1	1419	2	C	473	a
<i>OsRLCK91</i>	LOC_Os02g57700.1	2733	7	E	911	g1
<i>OsRLCK92</i>	LOC_Os02g58610.1	1095	0	C	365	a
<i>OsRLCK93</i>	LOC_Os03g01160.1	2352	8		784	g1
<i>OsRLCK94</i>	LOC_Os03g01410.1	1038	1		346	d
<i>OsRLCK95</i>	LOC_Os03g02190.1	1032	1	C	344	a
<i>OsRLCK96</i>	LOC_Os03g03880.1	1146	4	C	382	a

Table 1. Continued

Gene ID	Locus ID*	Size (bp)	Introns (No.)	EST/ cDNA*	Size (aa)	Domain [#]
<i>OsRLCK97</i>	LOC_Os03g03890.1	1173	4		391	a
<i>OsRLCK98</i>	LOC_Os03g04050.1	1596	8	C	532	a
<i>OsRLCK99</i>	LOC_Os03g05140.1	3093	0	E	1031	d
<i>OsRLCK100</i>	LOC_Os03g05470.1	1293	5	C	431	a
<i>OsRLCK101</i>	LOC_Os03g06330.1	1131	5	C	377	a
<i>OsRLCK102</i>	LOC_Os03g07430.1	1272	5	E	424	a
<i>OsRLCK103</i>	LOC_Os03g08170.1	1278	4	C	426	a
<i>OsRLCK104</i>	LOC_Os03g12520.1	1095	6	C	365	a
<i>OsRLCK105</i>	LOC_Os03g13820.1	1959	10	C	653	a
<i>OsRLCK106</i>	LOC_Os03g15770.1	1179	5	E	393	a
<i>OsRLCK107</i>	LOC_Os03g16740.1	1125	5	C	375	a
<i>OsRLCK108</i>	LOC_Os03g17550.1	1164	4	C	388	a
<i>OsRLCK109</i>	LOC_Os03g24930.1	1344	3	C	448	a
<i>OsRLCK110</i>	LOC_Os03g29410.1	1503	4	C	501	a
<i>OsRLCK111</i>	LOC_Os03g31000.1	2079	11		693	g1
<i>OsRLCK112</i>	LOC_Os03g31070.1	1557	5	C	519	g1
<i>OsRLCK113</i>	LOC_Os03g31260.1	570	2		190	a
<i>OsRLCK114</i>	LOC_Os03g44050.1	2262	2		754	f2
<i>OsRLCK115</i>	LOC_Os03g51040.1	1068	6		356	a
<i>OsRLCK116</i>	LOC_Os03g51060.1	438	1		146	a
<i>OsRLCK117</i>	LOC_Os03g56270.1	3090	1	C	1030	d
<i>OsRLCK118</i>	LOC_Os03g60710.1	1275	5	C	425	a
<i>OsRLCK119</i>	LOC_Os03g61010.1	1467	9	C	489	a
<i>OsRLCK120</i>	LOC_Os03g61060.1	1335	5	E	445	a
<i>OsRLCK121</i>	LOC_Os03g62180.1	2577	0	C	859	f4
<i>OsRLCK122</i>	LOC_Os03g62340.1	2817	8	C	939	h
<i>OsRLCK123</i>	LOC_Os03g62700.1	1107	6	C	369	a
<i>OsRLCK124</i>	LOC_Os04g01860.1	654	0		218	a
<i>OsRLCK125</i>	LOC_Os04g01874.1	2196	1	C	732	f1
<i>OsRLCK126</i>	LOC_Os04g03370.1	987	1		329	a
<i>OsRLCK127</i>	LOC_Os04g04330.1	3444	1	C	1148	d
<i>OsRLCK128</i>	LOC_Os04g04800.1	1020	1		340	a
<i>OsRLCK129</i>	LOC_Os04g06210.1	591	1		197	a
<i>OsRLCK130</i>	LOC_Os04g06244.1	1770	4		590	a
<i>OsRLCK131</i>	LOC_Os04g07200.1	1827	4		609	f5
<i>OsRLCK132</i>	LOC_Os04g09770.1	1071	1		357	a
<i>OsRLCK133</i>	LOC_Os04g12180.1	1128	4		376	a
<i>OsRLCK134</i>	LOC_Os04g12540.1	2814	3	E	938	f1
<i>OsRLCK135</i>	LOC_Os04g12580.1	2346	1		782	f1
<i>OsRLCK136</i>	LOC_Os04g12600.1	2415	0		805	f1
<i>OsRLCK137</i>	LOC_Os04g13640.1	975	1		325	a
<i>OsRLCK138</i>	LOC_Os04g15580.1	2502	6	C	834	f4
<i>OsRLCK139</i>	LOC_Os04g23620.1	2130	4		710	f1
<i>OsRLCK140</i>	LOC_Os04g24510.1	2052	4	E	684	f2
<i>OsRLCK141</i>	LOC_Os04g29740.1	1974	3		658	f2
<i>OsRLCK142</i>	LOC_Os04g29990.1	2124	2		708	f2
<i>OsRLCK143</i>	LOC_Os04g30010.1	2301	2	E	767	f2
<i>OsRLCK144</i>	LOC_Os04g30030.1	1392	5		464	j

Table 1. Continued

Gene ID	Locus ID*	Size (bp)	Introns (No.)	EST/ cDNA*	Size (aa)	Domain [#]
<i>OsRLCK145</i>	LOC_Os04g30040.1	2373	10		791	a
<i>OsRLCK146</i>	LOC_Os04g30060.1	1038	0		346	a
<i>OsRLCK147</i>	LOC_Os04g30260.1	2055	3		685	a
<i>OsRLCK148</i>	LOC_Os04g30270.1	1260	2		420	a
<i>OsRLCK149</i>	LOC_Os04g32310.1	1365	4	E	455	a
<i>OsRLCK150</i>	LOC_Os04g34300.1	2034	1		678	f5
<i>OsRLCK151</i>	LOC_Os04g35080.1	1581	6	C	527	a
<i>OsRLCK152</i>	LOC_Os04g41310.1	1101	6	E	367	a
<i>OsRLCK153</i>	LOC_Os04g42480.1	561	1	C	187	a
<i>OsRLCK154</i>	LOC_Os04g45730.1	1086	1	C	362	a
<i>OsRLCK155</i>	LOC_Os04g45920.1	1377	3	C	459	a
<i>OsRLCK156</i>	LOC_Os04g46320.1	1116	1	C	372	a
<i>OsRLCK157</i>	LOC_Os04g47620.1	1428	5	E	476	a
<i>OsRLCK158</i>	LOC_Os04g49460.1	969	0	C	323	a
<i>OsRLCK159</i>	LOC_Os04g52590.1	2793	19	C	931	a
<i>OsRLCK160</i>	LOC_Os04g52840.1	1119	5	C	373	a
<i>OsRLCK161</i>	LOC_Os04g52860.1	2535	0	C	845	a
<i>OsRLCK162</i>	LOC_Os04g53994.1	2442	6	C	814	f4
<i>OsRLCK163</i>	LOC_Os04g53998.1	2442	6	C	814	f4
<i>OsRLCK164</i>	LOC_Os04g54110.1	2064	7		688	f6
<i>OsRLCK165</i>	LOC_Os04g54120.1	2469	6		823	f5
<i>OsRLCK166</i>	LOC_Os04g54190.1	1140	5	E	380	a
<i>OsRLCK167</i>	LOC_Os04g56060.1	1176	5	C	392	a
<i>OsRLCK168</i>	LOC_Os04g56110.1	1293	1	E	431	a
<i>OsRLCK169</i>	LOC_Os04g56120.1	1296	1	C	432	a
<i>OsRLCK170</i>	LOC_Os04g56130.1	1260	1		420	a
<i>OsRLCK171</i>	LOC_Os04g56360.1	1185	5		395	a
<i>OsRLCK172</i>	LOC_Os04g58250.1	1125	5	C	375	a
<i>OsRLCK173</i>	LOC_Os04g58750.1	1077	8	C	359	a
<i>OsRLCK174</i>	LOC_Os04g58910.1	2817	1	C	939	d
<i>OsRLCK175</i>	LOC_Os05g01040.1	1191	7	C	397	a
<i>OsRLCK176</i>	LOC_Os05g02020.1	1188	5	C	396	a
<i>OsRLCK177</i>	LOC_Os05g03460.1	1332	4		444	a
<i>OsRLCK178</i>	LOC_Os05g04520.1	1086	6	C	362	a
<i>OsRLCK179</i>	LOC_Os05g07080.1	2160	4		720	f7
<i>OsRLCK180</i>	LOC_Os05g07470.1	1647	4		549	f4
<i>OsRLCK181</i>	LOC_Os05g12680.1	1416	7	E	472	a
<i>OsRLCK182</i>	LOC_Os05g14030.1	375	0		125	a
<i>OsRLCK183</i>	LOC_Os05g25370.1	2541	0	C	847	a
<i>OsRLCK184</i>	LOC_Os05g30820.1	912	5	E	304	a
<i>OsRLCK185</i>	LOC_Os05g30870.1	1476	4	C	492	a
<i>OsRLCK186</i>	LOC_Os05g32170.1	2040	0	C	680	a
<i>OsRLCK187</i>	LOC_Os05g38010.1	390	1		130	a
<i>OsRLCK188</i>	LOC_Os05g38770.1	1254	4	C	418	a
<i>OsRLCK189</i>	LOC_Os05g41950.1	1455	4	C	485	a
<i>OsRLCK190</i>	LOC_Os05g45010.1	1197	7		399	a
<i>OsRLCK191</i>	LOC_Os05g51190.1	1833	7		611	a
<i>OsRLCK192</i>	LOC_Os05g51400.1	1308	5		436	a

Table 1. Continued

Gene ID	Locus ID*	Size (bp)	Introns (No.)	EST/ cDNA ⁺	Size (aa)	Domain [#]
<i>OsRLCK193</i>	LOC_Os06g03610.1	2538	0	E	846	a
<i>OsRLCK194</i>	LOC_Os06g04880.1	2421	8	C	807	g1
<i>OsRLCK195</i>	LOC_Os06g05050.1	2268	4	E	756	f2
<i>OsRLCK196</i>	LOC_Os06g05830.1	1599	1	C	533	a
<i>OsRLCK197</i>	LOC_Os06g06760.1	2418	9	C	806	g4
<i>OsRLCK198</i>	LOC_Os06g06960.1	2418	0	E	806	f3
<i>OsRLCK199</i>	LOC_Os06g07070.1	1104	3	C	368	a
<i>OsRLCK200</i>	LOC_Os06g07230.1	1236	6	C	412	a
<i>OsRLCK201</i>	LOC_Os06g09230.1	1830	2	E	610	a
<i>OsRLCK202</i>	LOC_Os06g09860.1	2082	0	C	694	a
<i>OsRLCK203</i>	LOC_Os06g10160.1	1281	3	C	427	a
<i>OsRLCK204</i>	LOC_Os06g10230.1	2937	26	C	979	d
<i>OsRLCK205</i>	LOC_Os06g18810.1	2163	3		721	b
<i>OsRLCK206</i>	LOC_Os06g34960.1	1263	6	E	421	a
<i>OsRLCK207</i>	LOC_Os06g37620.1	2421	8	E	807	g2
<i>OsRLCK208</i>	LOC_Os06g38100.1	699	1		233	a
<i>OsRLCK209</i>	LOC_Os06g41500.1	1863	1		621	f1
<i>OsRLCK210</i>	LOC_Os06g44430.1	1179	3		393	a
<i>OsRLCK211</i>	LOC_Os06g45280.1	1410	5	C	470	a
<i>OsRLCK212</i>	LOC_Os06g45350.1	1668	7	C	556	a
<i>OsRLCK213</i>	LOC_Os06g46330.1	1149	5	C	383	a
<i>OsRLCK214</i>	LOC_Os06g47470.1	1290	6	E	430	a
<i>OsRLCK215</i>	LOC_Os06g47530.1	1668	8	E	556	a
<i>OsRLCK216</i>	LOC_Os06g47820.1	1179	5	C	393	a
<i>OsRLCK217</i>	LOC_Os06g48980.1	1209	4	C	403	a
<i>OsRLCK218</i>	LOC_Os06g50100.1	1146	3	C	382	a
<i>OsRLCK219</i>	LOC_Os06g50340.1	2985	1	C	995	d
<i>OsRLCK220</i>	LOC_Os06g51170.1	1245	5	C	415	a
<i>OsRLCK221</i>	LOC_Os07g03860.1	1965	1	E	655	a
<i>OsRLCK222</i>	LOC_Os07g04190.1	3072	1	E	1024	d
<i>OsRLCK223</i>	LOC_Os07g06570.1	918	6	E	306	a
<i>OsRLCK224</i>	LOC_Os07g07390.1	1149	3		383	f2
<i>OsRLCK225</i>	LOC_Os07g14490.1	990	1		330	a
<i>OsRLCK226</i>	LOC_Os07g15930.1	1722	3		574	a
<i>OsRLCK227</i>	LOC_Os07g26180.1	492	1		164	a
<i>OsRLCK228</i>	LOC_Os07g31130.1	1941	0	C	647	a
<i>OsRLCK229</i>	LOC_Os07g31190.1	2034	0	E	678	a
<i>OsRLCK230</i>	LOC_Os07g31210.1	2010	0		670	b
<i>OsRLCK231</i>	LOC_Os07g31250.1	1911	1	E	637	a
<i>OsRLCK232</i>	LOC_Os07g31290.1	1374	3		458	a
<i>OsRLCK233</i>	LOC_Os07g34980.1	1977	6		659	a
<i>OsRLCK234</i>	LOC_Os07g35280.1	2037	7	E	679	c
<i>OsRLCK235</i>	LOC_Os07g35390.1	1917	5	C	639	c
<i>OsRLCK236</i>	LOC_Os07g35630.1	1242	5		414	b
<i>OsRLCK237</i>	LOC_Os07g35730.1	753	3	E	251	a
<i>OsRLCK238</i>	LOC_Os07g35770.1	828	3		276	a
<i>OsRLCK239</i>	LOC_Os07g36980.1	864	3		288	a
<i>OsRLCK240</i>	LOC_Os07g47270.1	1440	4	C	480	a

Table 1. Continued

Gene ID	Locus ID*	Size (bp)	Introns (No.)	EST/ cDNA ⁺	Size (aa)	Domain [#]
<i>OsRLCK241</i>	LOC_Os07g48730.1	1056	4	E	352	a
<i>OsRLCK242</i>	LOC_Os07g49240.1	1503	5	C	501	a
<i>OsRLCK243</i>	LOC_Os07g49470.1	1173	4	C	391	a
<i>OsRLCK244</i>	LOC_Os08g03040.1	750	1		250	a
<i>OsRLCK245</i>	LOC_Os08g04420.1	1305	6	E	435	a
<i>OsRLCK246</i>	LOC_Os08g07890.1	4932	10	E	1644	a
<i>OsRLCK247</i>	LOC_Os08g10070.1	1110	5	C	370	a
<i>OsRLCK248</i>	LOC_Os08g13870.1	2373	0		791	f1
<i>OsRLCK249</i>	LOC_Os08g15060.1	1806	10	C	602	h
<i>OsRLCK250</i>	LOC_Os08g24630.1	912	0	E	304	a
<i>OsRLCK251</i>	LOC_Os08g25430.1	1776	8	C	592	a
<i>OsRLCK252</i>	LOC_Os08g27780.1	1038	0		346	a
<i>OsRLCK253</i>	LOC_Os08g28710.1	1464	1	C	488	a
<i>OsRLCK254</i>	LOC_Os08g29040.1	1149	2		383	a
<i>OsRLCK255</i>	LOC_Os08g35600.1	1437	3	C	479	a
<i>OsRLCK256</i>	LOC_Os08g38560.1	3315	1	C	1105	d
<i>OsRLCK257</i>	LOC_Os08g39590.1	1974	1	E	658	d
<i>OsRLCK258</i>	LOC_Os08g40990.1	2154	1	C	718	d
<i>OsRLCK259</i>	LOC_Os09g03620.1	1581	6	C	527	a
<i>OsRLCK260</i>	LOC_Os09g12300.1	525	1		175	a
<i>OsRLCK261</i>	LOC_Os09g15580.1	1485	9		495	a
<i>OsRLCK262</i>	LOC_Os09g16590.1	1002	1		334	a
<i>OsRLCK263</i>	LOC_Os09g17890.1	981	6		327	b
<i>OsRLCK264</i>	LOC_Os09g17910.1	2415	11		805	d
<i>OsRLCK265</i>	LOC_Os09g18230.1	2586	11	E	862	a
<i>OsRLCK266</i>	LOC_Os09g19140.1	2526	11		842	a
<i>OsRLCK267</i>	LOC_Os09g19380.1	801	3		267	a
<i>OsRLCK268</i>	LOC_Os09g19700.1	1539	5	C	513	a
<i>OsRLCK269</i>	LOC_Os09g20880.1	1083	5	C	361	a
<i>OsRLCK270</i>	LOC_Os09g24330.1	528	1		176	a
<i>OsRLCK271</i>	LOC_Os09g24340.1	903	1	E	301	a
<i>OsRLCK272</i>	LOC_Os09g27010.1	1377	3	C	459	a
<i>OsRLCK273</i>	LOC_Os09g29520.1	1032	0	C	344	a
<i>OsRLCK274</i>	LOC_Os09g29540.1	765	0	C	255	a
<i>OsRLCK275</i>	LOC_Os09g30190.1	3348	1	E	1116	d
<i>OsRLCK276</i>	LOC_Os09g33630.1	864	7	C	288	a
<i>OsRLCK277</i>	LOC_Os09g33860.1	1086	6	E	362	a
<i>OsRLCK278</i>	LOC_Os09g36320.1	1221	3	C	407	a
<i>OsRLCK279</i>	LOC_Os09g37800.1	2568	6		856	f1
<i>OsRLCK280</i>	LOC_Os09g39620.1	2577	8		859	g1
<i>OsRLCK281</i>	LOC_Os09g39640.1	2193	7	C	731	g1
<i>OsRLCK282</i>	LOC_Os09g39650.1	1848	8	C	616	a
<i>OsRLCK283</i>	LOC_Os09g39860.1	1068	0	C	356	a
<i>OsRLCK284</i>	LOC_Os09g39930.1	1260	1	C	420	a
<i>OsRLCK285</i>	LOC_Os10g01060.1	2436	7	C	812	g1
<i>OsRLCK286</i>	LOC_Os10g01390.1	948	4		316	a
<i>OsRLCK287</i>	LOC_Os10g02360.1	1935	0	C	645	b
<i>OsRLCK288</i>	LOC_Os10g02990.1	1887	4		629	d

Table 1. Continued

Gene ID	Locus ID*	Size (bp)	Introns (No.)	EST/ cDNA*	Size (aa)	Domain#
<i>OsRLCK289</i>	LOC_Os10g03990.1	879	2		293	a
<i>OsRLCK290</i>	LOC_Os10g05160.1	1968	4		656	a
<i>OsRLCK291</i>	LOC_Os10g05170.1	483	0		161	a
<i>OsRLCK292</i>	LOC_Os10g05250.1	1953	0	E	651	b
<i>OsRLCK293</i>	LOC_Os10g05400.1	1953	0		651	b
<i>OsRLCK294</i>	LOC_Os10g10870.1	861	0	C	287	a
<i>OsRLCK295</i>	LOC_Os10g12620.1	2388	6	C	796	i
<i>OsRLCK296</i>	LOC_Os10g18260.1	1095	5	E	365	a
<i>OsRLCK297</i>	LOC_Os10g24850.1	375	0		125	a
<i>OsRLCK298</i>	LOC_Os10g25550.1	1170	4	C	390	a
<i>OsRLCK299</i>	LOC_Os10g26520.1	1287	5	C	429	a
<i>OsRLCK300</i>	LOC_Os10g29620.1	1143	4	C	381	a
<i>OsRLCK301</i>	LOC_Os10g30600.1	1386	4	E	462	a
<i>OsRLCK302</i>	LOC_Os10g34220.1	861	3	C	287	a
<i>OsRLCK303</i>	LOC_Os10g37190.1	1098	1	E	366	a
<i>OsRLCK304</i>	LOC_Os10g39670.1	1569	8	C	523	a
<i>OsRLCK305</i>	LOC_Os10g40060.1	2067	7	C	689	g1
<i>OsRLCK306</i>	LOC_Os10g40100.1	2700	8	C	900	g1
<i>OsRLCK307</i>	LOC_Os10g41220.1	2349	8	C	783	g1
<i>OsRLCK308</i>	LOC_Os10g42110.1	1539	9	C	513	a
<i>OsRLCK309</i>	LOC_Os11g02830.1	1242	5	C	414	a
<i>OsRLCK310</i>	LOC_Os11g03880.1	2487	0	E	829	f8
<i>OsRLCK311</i>	LOC_Os11g06780.1	1167	1	C	389	a
<i>OsRLCK312</i>	LOC_Os11g07130.1	1344	2		448	a
<i>OsRLCK313</i>	LOC_Os11g07190.1	801	3		267	a
<i>OsRLCK314</i>	LOC_Os11g07240.1	2643	4	E	881	d
<i>OsRLCK315</i>	LOC_Os11g08950.1	1626	7	C	542	a
<i>OsRLCK316</i>	LOC_Os11g10640.1	2148	14	E	716	e
<i>OsRLCK317</i>	LOC_Os11g10690.1	1227	8	E	409	a
<i>OsRLCK318</i>	LOC_Os11g10710.1	1959	13	E	653	e
<i>OsRLCK319</i>	LOC_Os11g11780.1	807	4	C	269	a
<i>OsRLCK320</i>	LOC_Os11g12620.1	3186	1	E	1062	d
<i>OsRLCK321</i>	LOC_Os11g17380.1	2082	7		694	j
<i>OsRLCK322</i>	LOC_Os11g17440.1	3108	18	E	1036	l
<i>OsRLCK323</i>	LOC_Os11g18670.1	1062	5		354	i
<i>OsRLCK324</i>	LOC_Os11g18980.1	1401	4		467	a
<i>OsRLCK325</i>	LOC_Os11g19490.1	897	1	C	299	a
<i>OsRLCK326</i>	LOC_Os11g25510.1	1140	0	C	380	a
<i>OsRLCK327</i>	LOC_Os11g25860.1	2598	0	C	866	b
<i>OsRLCK328</i>	LOC_Os11g29510.1	924	6		308	a
<i>OsRLCK329</i>	LOC_Os11g34610.1	1005	5		335	a
<i>OsRLCK330</i>	LOC_Os11g34624.1	1791	6		597	m
<i>OsRLCK331</i>	LOC_Os11g35100.1	705	4		235	a
<i>OsRLCK332</i>	LOC_Os11g35120.1	2040	0		680	a
<i>OsRLCK333</i>	LOC_Os11g35260.1	678	0		226	a
<i>OsRLCK334</i>	LOC_Os11g35274.1	1929	0	E	643	b
<i>OsRLCK335</i>	LOC_Os11g35290.1	1986	0		662	a
<i>OsRLCK336</i>	LOC_Os11g39420.1	1977	10	E	659	a

Table 1. Continued

Gene ID	Locus ID*	Size (bp)	Introns (No.)	EST/ cDNA*	Size (aa)	Domain#
<i>OsRLCK337</i>	LOC_Os11g39450.1	1371	5	E	457	a
<i>OsRLCK338</i>	LOC_Os11g39490.1	2514	11		838	k
<i>OsRLCK339</i>	LOC_Os11g39530.1	1698	8	E	566	a
<i>OsRLCK340</i>	LOC_Os11g40380.1	2151	9		717	a
<i>OsRLCK341</i>	LOC_Os11g40430.1	1416	3		472	a
<i>OsRLCK342</i>	LOC_Os11g42440.1	1812	7		604	a
<i>OsRLCK343</i>	LOC_Os11g44250.1	1632	1	E	544	a
<i>OsRLCK344</i>	LOC_Os11g44260.1	1086	2		362	a
<i>OsRLCK345</i>	LOC_Os11g44430.1	1614	1	E	538	a
<i>OsRLCK346</i>	LOC_Os11g44500.1	1743	1	E	581	a
<i>OsRLCK347</i>	LOC_Os11g44550.1	918	1		306	a
<i>OsRLCK348</i>	LOC_Os11g44560.1	2982	7	C	994	b
<i>OsRLCK349</i>	LOC_Os11g44660.1	1371	6		457	a
<i>OsRLCK350</i>	LOC_Os11g44690.1	1440	8		480	a
<i>OsRLCK351</i>	LOC_Os11g44860.1	1452	5	C	484	a
<i>OsRLCK352</i>	LOC_Os11g45540.1	1878	5		626	c
<i>OsRLCK353</i>	LOC_Os11g46870.1	2040	3		680	f2
<i>OsRLCK354</i>	LOC_Os11g46960.1	3093	3	E	1031	d
<i>OsRLCK355</i>	LOC_Os11g47000.1	3006	4	E	1002	d
<i>OsRLCK356</i>	LOC_Os11g47170.1	576	1		192	a
<i>OsRLCK357</i>	LOC_Os11g47490.1	456	3		152	a
<i>OsRLCK358</i>	LOC_Os12g01910.1	2664	3		888	n
<i>OsRLCK359</i>	LOC_Os12g02810.1	1110	4	C	370	a
<i>OsRLCK360</i>	LOC_Os12g03690.1	2487	0		829	f8
<i>OsRLCK361</i>	LOC_Os12g07990.1	2217	7		739	h
<i>OsRLCK362</i>	LOC_Os12g08050.1	924	4		308	a
<i>OsRLCK363</i>	LOC_Os12g13310.1	891	1		297	a
<i>OsRLCK364</i>	LOC_Os12g14480.1	1095	4		365	a
<i>OsRLCK365</i>	LOC_Os12g14610.1	1458	6		486	a
<i>OsRLCK366</i>	LOC_Os12g14699.1	1965	8	C	655	a
<i>OsRLCK367</i>	LOC_Os12g16520.1	1530	1	E	510	a
<i>OsRLCK368</i>	LOC_Os12g29434.1	978	2		326	a
<i>OsRLCK369</i>	LOC_Os12g30180.1	1437	6	C	479	a
<i>OsRLCK370</i>	LOC_Os12g31560.1	1569	1		523	a
<i>OsRLCK371</i>	LOC_Os12g31610.1	1392	5		464	a
<i>OsRLCK372</i>	LOC_Os12g33600.1	390	0		130	a
<i>OsRLCK373</i>	LOC_Os12g41710.1	1137	5	E	379	a
<i>OsRLCK374</i>	LOC_Os12g42040.1	2226	2	E	742	f2
<i>OsRLCK375</i>	LOC_Os12g42070.1	2196	2	C	732	f2
<i>OsRLCK376</i>	AK066081	1977		C	334	a
<i>OsRLCK377</i>	AK070591	1566		C	228	a
<i>OsRLCK378</i>	AK105147	570		C	118	a
<i>OsRLCK379</i>	AK119388	2776		C	731	f9

* Chromosomal location is shown after Loc_Os under Locus ID;
+ C- full length cDNA, E- EST;
Domain organization refers to Figure 1.

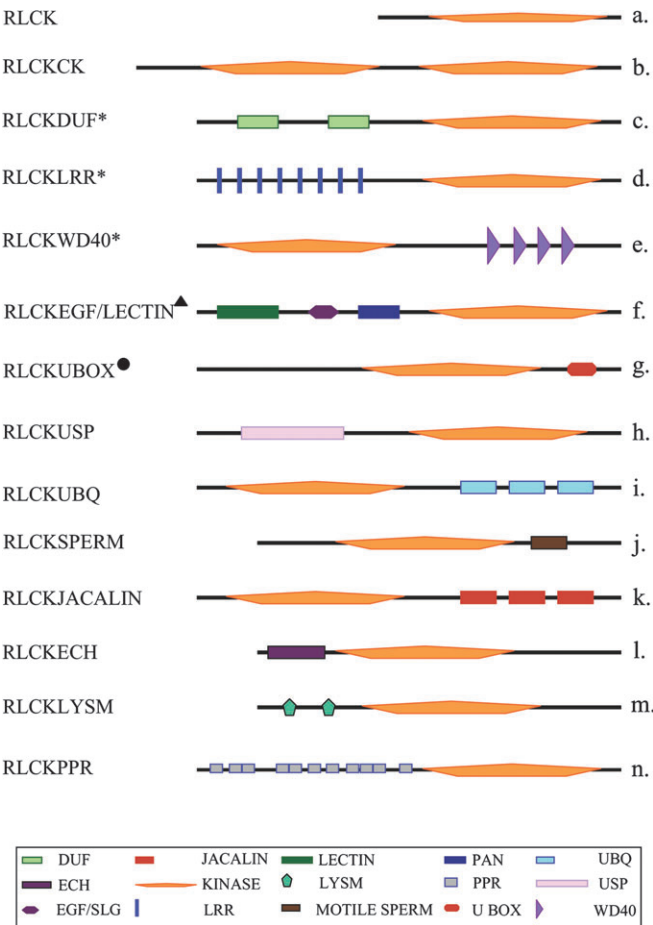


Figure 1. Domain Organization of the *OsRLCK* Protein Family. The SMART (<http://smart.embl-heidelberg.de/>) database was used to obtain the details of domain organization. Fourteen major types of domain organizations include RLCK (275), RLCKCK (10), RLCKDUF (4), RLCKLRR (23), RLCKWD40 (2), RLCKEGF/LECTIN (36), RLCKUBOX (15), RLCKUSP (6), RLCKUBQ (2), RLCKSPERM (2), RLCKJACALIN (1), RLCKECH (1), RLCKLYSM (1), and RLCKPPR (1). The asterisk on RLCKDUF, RLCKLRR, and RLCKWD40 indicates that the number and relative position of domains, namely DUF, LRR, and WD40, respectively, in these three categories of proteins could be variable. The filled circle on RLCKUBOX indicates that this domain is occasionally seen with some additional domains such as TPR and USP. The triangle mark on RLCKEGF/LECTIN is used to indicate that the domain organization for this class is only representative, since this particular class is much more diverse. Of the 36 RLCKEGF/LECTIN members, the actual organization is kinase+EGF (9), kinase+EGF+EGF (12), kinase+HDAC+Lectin (3), kinase+LECTIN (5), kinase+PAN+SLG+LECTIN (3), kinase+EGF+LECTIN (1), kinase+APPL+LECTIN (1), kinase+PAN+LECTIN (1), and kinase+PAN+EGF+LECTIN (1).

Table 2). Association of the *OsRLCKs* to abiotic stress-related QTLs was analyzed by using the Gramene database (www.gramene.org/qtl/index.html). Of a total of 52 traits available in this category, the *OsRLCK* gene family showed associa-

tion with 38 QTLs belonging to abiotic stress-related traits (Figure 2).

Phylogenetic Analyses of the *OsRLCK* Family

For *OsRLCKs*, a total of 58 clades could be identified with $\geq 50\%$ bootstrap support, which included six major clades (with at least 10 members; Figure 3). Yeast kinases and rice RLCKs formed well separated clades. Interestingly, however, seven yeast kinases grouped with rice RLCKs suggesting that these could represent ancient lineage that could be involved in basic cell functioning (Supplemental Figure 1).

Expression Profiles of *OsRLCKs* during Development

To analyze the genome-wide expression profiles of rice *OsRLCKs*, microarray analysis was carried out using Affymetrix rice whole genome array. A total of 323 probeset IDs corresponding to rice *OsRLCKs* were found on the Affymetrix gene chip. In addition, evidence for transcription activity for 16 more *OsRLCKs* was obtained from the MPSS database (<http://mpss.udel.edu/rice/>) (Supplemental Table 3). Hence, expression profiles of 339 *OsRLCKs* could be analyzed.

OsRLCKs expression was analyzed during entire reproductive development comprising six panicles (P1–P6) and five seed (S1–S5) developmental stages and compared with vegetative stages comprising mature leaf, root, and seedling. A total of 98 *OsRLCKs* were found to express differentially during reproductive developmental stages. Of these, 44 and 18 were up-regulated during stages of panicle and seed development, respectively, out of which eight *OsRLCKs* are common to both reproductive stages. Overall, 15 *OsRLCKs* are commonly down-regulated out of 33 and 32 *OsRLCKs*, down-regulated in panicle and seed, respectively (Figure 4). Based on the similarity in expression profiles, the differentially expressed *OsRLCKs* were categorized into five major groups, A–E (Figure 5). Group A represents the genes that are up-regulated both in panicle and seed developmental stages. The accumulation of their transcripts is relatively higher in panicle. Three group A members (*OsRLCK85*, 204, and 257) were found to be highly up-regulated in panicle, having the highest expression in P1 (935 folds change for *OsRLCK204* and ~ 100 folds change for *OsRLCK85* and 257) followed by a gradual decrease in the subsequent stages (Supplemental Table 4). Genes up-regulated only in panicle developmental stages are categorized as group B. Most of the members of this group expressed in the initial panicle developmental stages. Group C comprised three genes (*OsRLCK82*, 98, and 243), which expressed during late seed developmental stages (S4 and S5). There are two genes (*OsRLCK539* and 241) in group D, displaying transcript accumulation at late panicle

developmental stages (P6). Group E consists of genes down-regulated in both panicle and seed developmental stages. Of these, *OsRLCK120*, *122*, *203*, *220*, *240*, and *334* were uniformly down-regulated throughout the reproductive stages. Expression evidences from MPSS signature data identified five additional *OsRLCKs* showing high expression during reproductive stages and seven *OsRLCKs* having marginal to high expression during vegetative stages (Supplemental Table 3).

Expression Profiles of *OsRLCKs* during Abiotic and Biotic Stress Conditions

The expression profiles of *OsRLCKs* in rice seedling were analyzed under abiotic stress conditions (cold, salt, and dehydration). Out of 82 differentially expressed *OsRLCKs*, 46 were up-regulated and 36 were down-regulated under any one of the three abiotic stress conditions. Based on the expression pattern, genes are grouped into six groups (A–F; Figure 6). Five *OsRLCKs* (*72*, *101*, *106*, *154*, and *272*) were found to be up-regulated, whereas three *OsRLCKs* (*95*, *308*, and *375*) were down-regulated in all three stress conditions. *OsRLCKs* specifically down- or up-regulated in desiccation stress were categorized as groups B and D, respectively, whereas group C members induced under salt stress. Additionally, eight more *OsRLCKs* were found to be abiotic stress-responsive as revealed by MPSS signatures (Supplemental Table 3). Out of a total of 90 abiotic stress-responsive genes, 62 genes were found to be associated with one or other abiotic stress QTLs (Figure 2).

Keeping in view the observations that certain expression patterns during stress and reproduction overlap (Ray et al., 2007), differentially expressed *OsRLCKs* during any developmental stage or abiotic stress condition were analyzed (Supplemental Table 4). Among the differentially expressed *OsRLCKs*, 16 abiotic stress-responsive genes were found to be commonly up-regulated with reproductive development; on other hand, 10 *OsRLCKs* were commonly down-regulated under abiotic stress and stages of reproductive development (Figure 4). It was also found that four *OsRLCKs* up-regulated during salt, dehydration, and/or cold stress were down-regulated in all the panicle developmental stages. Moreover, six *OsRLCKs* down-regulated in salt and dehydration stress were up-regulated throughout the panicle development stages (Supplemental Table 4).

Expression pattern of the 16 genes differentially expressed during three abiotic stresses and representing different domain compositions were confirmed by real-time PCR. Out of 48 samples tested, 44 showed similar expression patterns representing most of the genes and all three stresses (Figure 7).

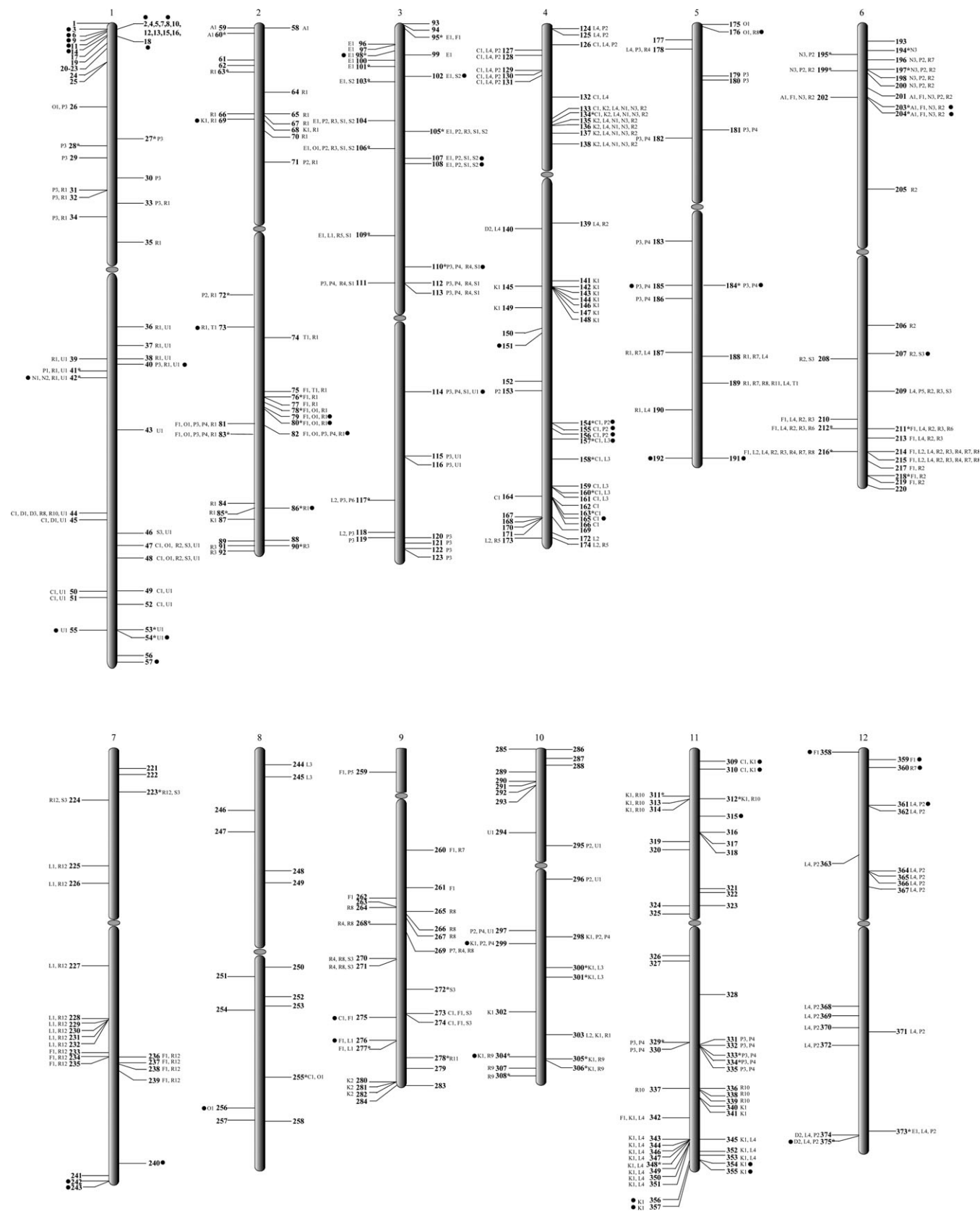
Evidence for expression of *OsRLCKs* in response to biotic stress was derived from publicly available MPSS libraries at the <http://mpss.udel.edu/rice> database. A total of 120 genes

showed significant (≥ 3 TPM) differential expression either in pathogen-infected (*Xanthomonas oryzae* and *Magnaporthe grisea*) or mechanically wounded samples (Supplemental Table 5). Out of 120 differentially expressing genes, 89 were found to be at least two-fold up-regulated in at least one of the treatments, while 31 were at least two-fold down-regulated in at least one of the treatments (but not up-regulated at least two-fold in any one treatment) in relation to untreated control plants (Supplemental Figure 2). Interestingly, resistant transgenic and sensitive untransformed plants showed quantitative as well as qualitative differences in the level of gene induction during these treatments (Supplemental Figure 2). Furthermore, ~50% biotic stress-related *OsRLCKs* belong to four out of the six major phylogenetic clades (2, 34, 35, and 58) and 68% members of clade 34 showed biotic stress-responsiveness (Figure 3). We have also analyzed *OsRLCKs* for domain class (Figure 1) specific expression patterns; however, members of different domain organizations showed different expression patterns.

In comparison with abiotic stress-related *OsRLCKs*, 45 were found to be common between these two types of stresses. Among them, 20 genes were commonly up-regulated, while only three genes were down-regulated in both types of stresses. Specific up-regulation in one type of stress and down-regulation in the other were also observed for 22 genes (Supplemental Figure 2).

Expression Profiling of Duplicated *OsRLCKs*

Expression profiles for 19 *OsRLCK* pairs from segmentally duplicated regions were analyzed for which probset IDs could be assigned on Affymetrix whole genome array chip. The average signal values for all the sample were presented as an area-diagram (Supplemental Figure 3A). Seven duplicated sets showed subfunctionalization while neofunctionalization was observed in five duplicated pairs. Out of the remaining, there were eight cases wherein one of the members had negligible expression levels displaying pseudofunctionalization. For 49 tandemly duplicated *OsRLCKs*, belonging to 21 groups, comparative expression analysis was also done. Out of these, probe sets were available for 29 genes, including all the members of 11 groups and more than one member of three groups. Thus, out of 21 groups of tandemly duplicated *OsRLCKs*, expression patterns could be analyzed for 14 groups. Duplicated *OsRLCKs* belonging to three groups (5, 11, and 14) may retain their original function as indicated by similar expression patterns. Neofunctionalization was evident in members of five groups (2, 4, 17, 18, and 19) while, members of two groups (12 and 15) displayed subfunctionalization. Pseudofunctionalization was seen in members of four groups (1, 3, 6, and 16) (Supplemental Figure 3B). The fate of expression of duplicated *OsRLCK*

Figure 2. Chromosomal Localization of the *OsRLCK* Gene Family Members.

genes was compared between segmental and tandem duplication events. Almost the same trends of functionalization were seen between two types of duplication events. However, none of the segmentally duplicated *OsRLCK* genes could retain its original expression pattern (Figure 8).

DISCUSSION

RLKs represent one of the largest families of proteins in plants with diverse biological roles, which include hormonal response, cell differentiation, plant growth and development, self-incompatibility, stress response, symbiont and pathogen recognition (Morris and Walker, 2003; Morillo and Tax, 2006). About 25% of RLKs in the *Arabidopsis* genome lack transmembrane domain and are referred to as RLCKs (Shiu and Bleeker, 2001a). Previously, 187 RLCKs were identified in the *indica* rice genome (Shiu et al., 2004). However, we have presented evidence for the existence of 379 *OsRLCKs* in *japonica* rice. The difference in number could be due to the use of an older version of the draft sequence of *indica* rice (Yu et al., 2002) used for analysis by Shiu et al. (2004), sub-species-specific difference at the sequence level or the approach used for analysis (International Rice Genome Sequencing Project, 2005M; Vij et al., 2006). In fact, we could identify a much larger number of RLCKs (300) in *indica* genome upon searching the improved version (Yu et al., 2005). The number of *OsRLCKs* (379) is also twice that of RLCKs reported from *Arabidopsis* (147). It is possible that this variability in number is a result of independent lineages in the monocots and dicots. Shiu et al. (2004) had estimated the number of RLKs before *Arabidopsis*/rice divergence to be ~443, suggesting that the RLK family was large even in the common ancestors of flowering plants. Evolutionary studies have shown that large gene families such as that of RLKs are usually an exception and the retention of duplicates, especially in plants, represents a plant-specific evolution and are needed for extra-cellular signal sensing and propagation. In fact, RLKs involved in defense/disease resistance have undergone considerable expansion in relation to other gene families (Shiu et al., 2004; Morillo

and Tax, 2006). In our analysis also, the expression of ~32% genes of *OsRLCK* genes was modulated in response to biotic stress(es).

Domain analyses for the RLCKs showed that the majority had only the cytoplasmic kinase domain (more than 70%). Among the additional domains, the majority were leucine-rich repeats (LRRs) and carbohydrate-binding motifs (e.g. LysM, PAN, and lectin), which is also the case with the receptor domains found in typical RLKs (Shiu et al., 2004). These domains, along with other domains present in RLCKs, are involved in protein-protein interaction, Ca^{2+} binding, glycoprotein binding, recognition of pathogen cell wall components, hormone signaling, and some have unknown functions (Shiu and Bleeker, 2001b). Database search for similar RLKs in primitive eukaryotes showed that representatives of RLKs were absent from yeast and *Neurospora*. However, several RLKs (TKLs) and RLCKs (IRE family members) were found in the slime mould (*Dictyostelium discoideum*; Goldberg et al., 2006), indicating that the RLCKs present in plants and animals probably had an early ancestral form. A bryophyte, *Physcomitrella*, encodes only 157 receptor-like kinases (www.cosmos.org/bm/isearch; Rensing et al., 2008) reflecting on possible lineage-specific expansion in higher plants. Furthermore, cytoplasmic RLKs from animals such as Pelle-like and raf kinases were seen to interact with cell surface receptors and transduce the signal to downstream components. This suggests that RLCKs in plants may represent a group formed by ancestral cytoplasmic kinases as well as RLCKs derived from RLKs by the loss of transmembrane domain. In such a scenario, RLCKs may form a complex with RLPs, since chimeric RLK carrying receptor domain of BRI1 and kinase domain of Xa21 are able to transduce the signal (He et al., 2000). Further, RLPs alone could also perform the function of RLKs (Wang et al., 1998; Trotochaud et al., 1999) possibly by interacting with RLK/RLCKs. This specialized function of cytoplasmic RLCKs would probably explain their retention in higher plants and animals (Shiu and Bleeker, 2001a).

To evaluate the relationship between phylogenetically related *OsRLCKs* and their functions, six distinct clades were selected with a bootstrap value of $\geq 50\%$ and with a minimum of 10 genes. Clade 34 has the maximum number of 54 genes,

The genes are numbered in ascending order of their positions on the 12 rice chromosomes (refer to Table 1 for details). Filled circles indicate genes present in duplicated segments. Details are given in Supplemental Table 1. The association of genes falling within limits of abiotic stress-related QTLs available on the Gramene database (www.gramene.org) is shown. QTLs for the various traits are shown as follows; aluminum sensitivity (A1), cold tolerance (C1), deep root to shoot ratio (D1), drought susceptibility index (D2), elongation ability (E1), iron sensitivity (F1), KClO₃ resistance (K1), potassium uptake (K2), leaf rolling (L1), leaf drying (L2), leaf rolling time (L3), lodging incidence (L4), sodium to potassium ratio (N1), sodium uptake (N2), sodium concentration (N3), osmotic adjustment capacity (O1), plant survival percentage under submergence (P1), phosphorus sensitivity (P2), penetrated root number (P3), penetrated to total root ratio (P4), penetrated root thickness (P5), relative phosphorus distribution between shoot and root, penetrated root length (P6) (R1), relative phosphorus concentration (R2), root penetration index (R3), relative growth rate (R4), root pulling force (R5), relative phosphorus uptake (R6), relative water content (R7), root weight (R8), relative phosphorus utilization efficiency (R9), rooting depth (R10), relative root length (R11), stomatal closure time (S1), stomatal resistance (S2), salt sensitivity (S3), total shoot elongation under submergence (T1), and ultraviolet-b resistance (U1). The differentially expressed genes during abiotic stress and falling in relevant QTL regions are also marked (*).

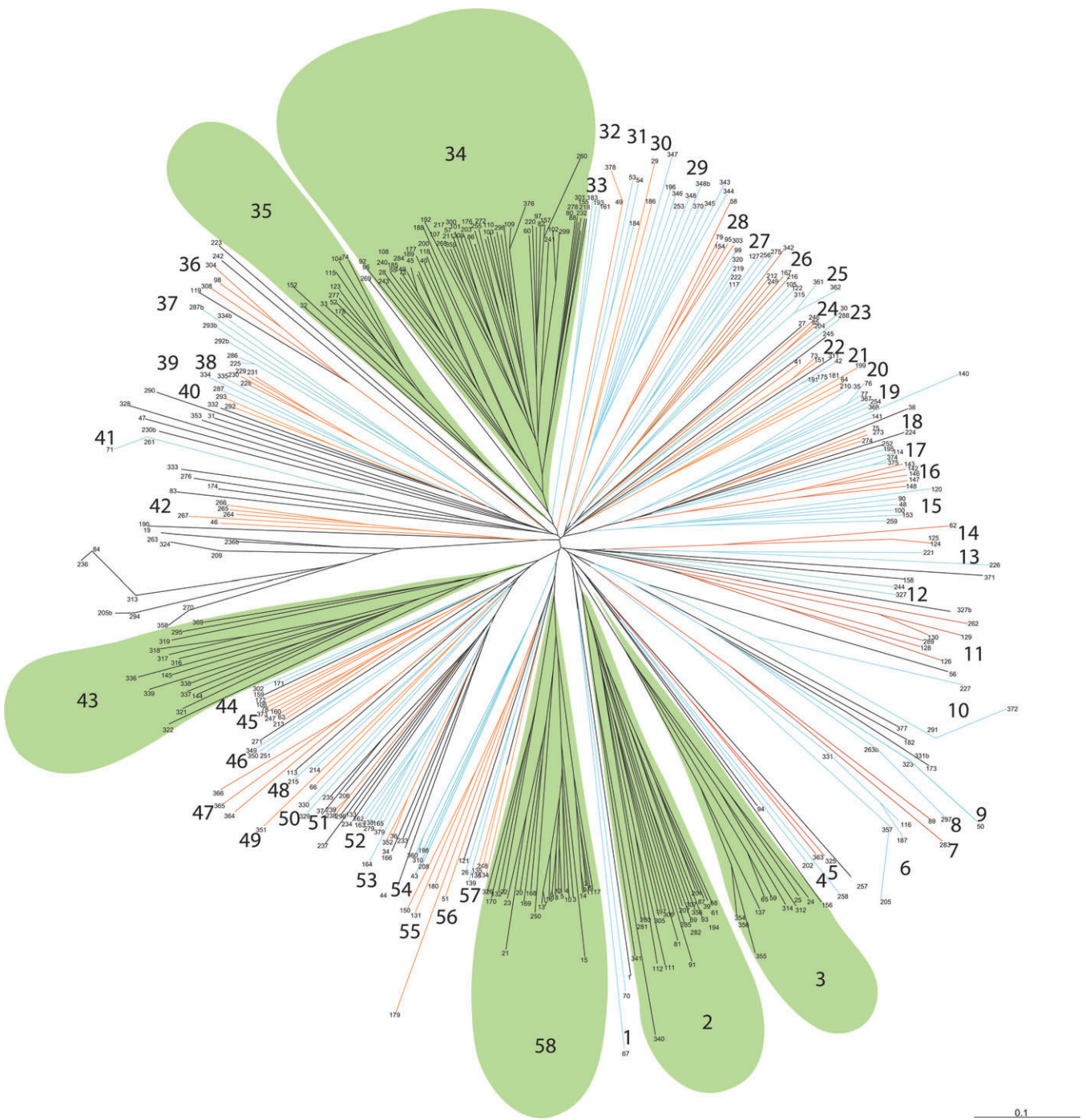


Figure 3. Phylogenetic Relationships amongst the Members of the *OsRLCK* Gene Family. The neighbor joining tree was generated using ClustalX for the 379 *OsRLCK* protein sequences and viewed using Treeview. Scale bar represents 0.1 amino acid substitutions per site. Based on bootstrap values, the proteins have been classified into 58 clades marked with red and blue colored lines. The six major clades (with ≥ 10 members each) are shaded in green.

out of which 38 were differentially expressed during reproductive development and/or in response to abiotic stress. Similarly, four clades were found to be enriched with biotic stress-responsive genes; clade 34 contains ~68% of

such genes. Although no generalization could be made for clade-specific functionalization, however, it seems that biotic stress-related *OsRLCK*s have some protein level conservation.

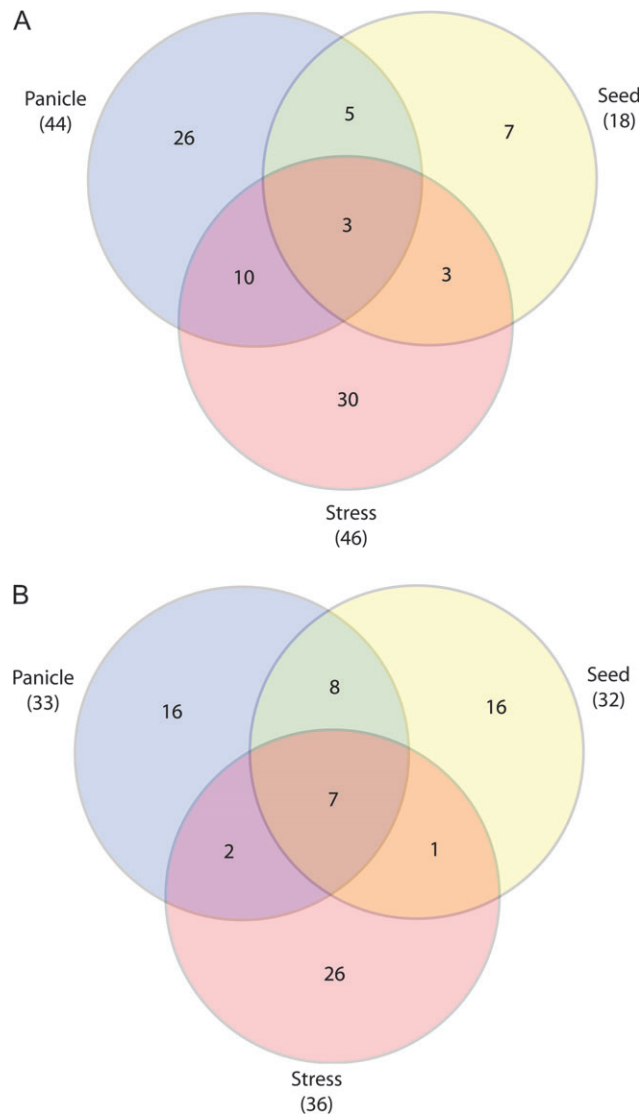


Figure 4. Venn Diagram for Differentially Expressed *OsRLCK*s during Selected Reproductive Developmental Stages (Panicle and Seed) and under Abiotic Stress Conditions (Cold, Salt and Dehydration Stress).

(A) Up-regulated *OsRLCK*s.

(B) Down-regulated *OsRLCK*s. A gene is considered differentially expressed if it is up- or down-regulated at least two-fold, at P -value < 0.05 , with respect to all vegetative stages (seedling, mature leaf, and root) in the case of reproductive development and 7-day-old seedling for the stress sample.

A number of *RLCK* genes (108) found in the segmental and tandemly duplicated regions of the rice genome suggest a significant role of chromosome segment/gene duplications in evolution of this gene family. Duplicated *OsRLCK* genes showed different fate in terms of expression patterns, probably because of lack of intense evolutionary selection pressure and need for diversification (Lynch and

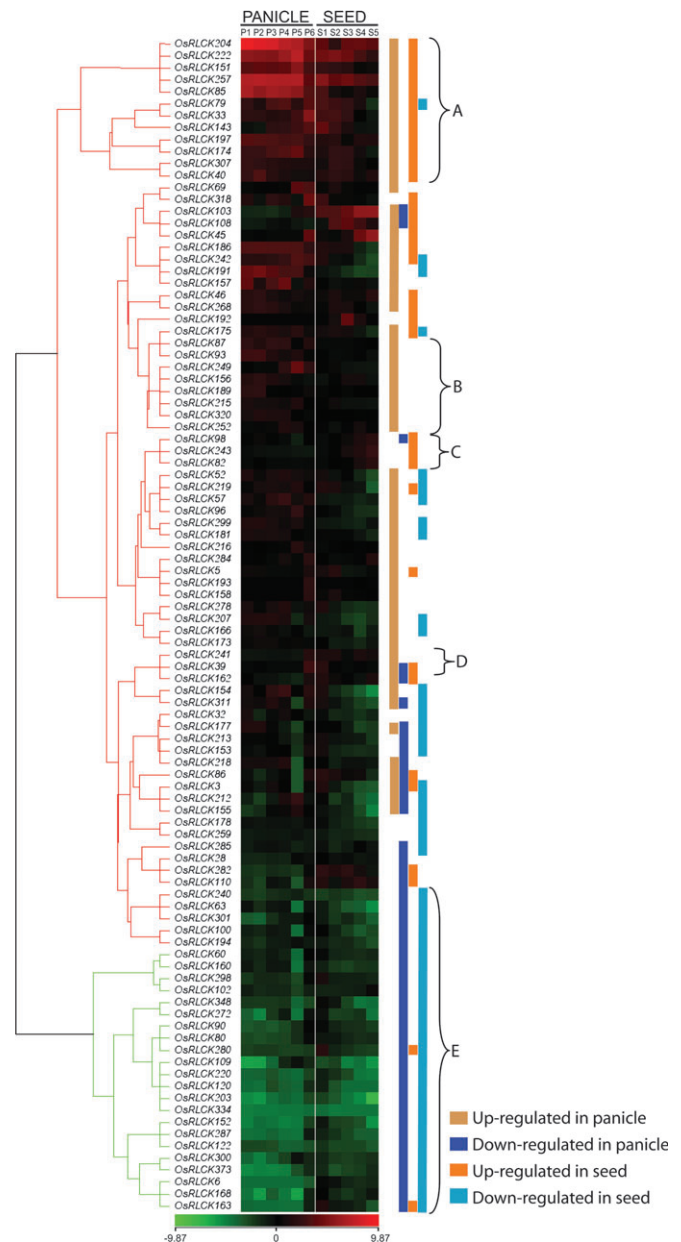


Figure 5. Expression Profiles of *OsRLCK*s under Reproductive Development Comprising Six Stages of Panicle (P1 (0–3 cm), P2 (3–5 cm), P3 (5–10 cm), P4 (10–15 cm), P5 (15–22 cm), and P6 (22–30 cm)) and Five Stages of Seed ((S1 (0–2 DAP), S2 (3–4 DAP), S3 (4–10 DAP), and S4 (10–15 DAP), and S5 (15–20 DAP)) Development).

A gene is considered differentially expressed during reproductive development if it is up- or down-regulated at least two-fold, at P -value < 0.05 , with respect to all the vegetative controls, viz. 7-day-old seedling, mature leaf, and root. Clustering of the expression profile was done with log transformed average values taking mature leaf as base-line. The color scale at the bottom of the heat map is given in Log₂ intensity value. The vertical color bars on the right represent genes that are differentially regulated more than two-fold in any one stage of panicle or seed development with reference to mature leaf. Based on the expression pattern, genes were grouped into five categories: A, up-regulated both in panicle and seed; B, up-regulated in panicle; C, up-regulated in seed; D, up-regulated in late panicle developmental stage (P6); E, down-regulated in both panicle and seed.

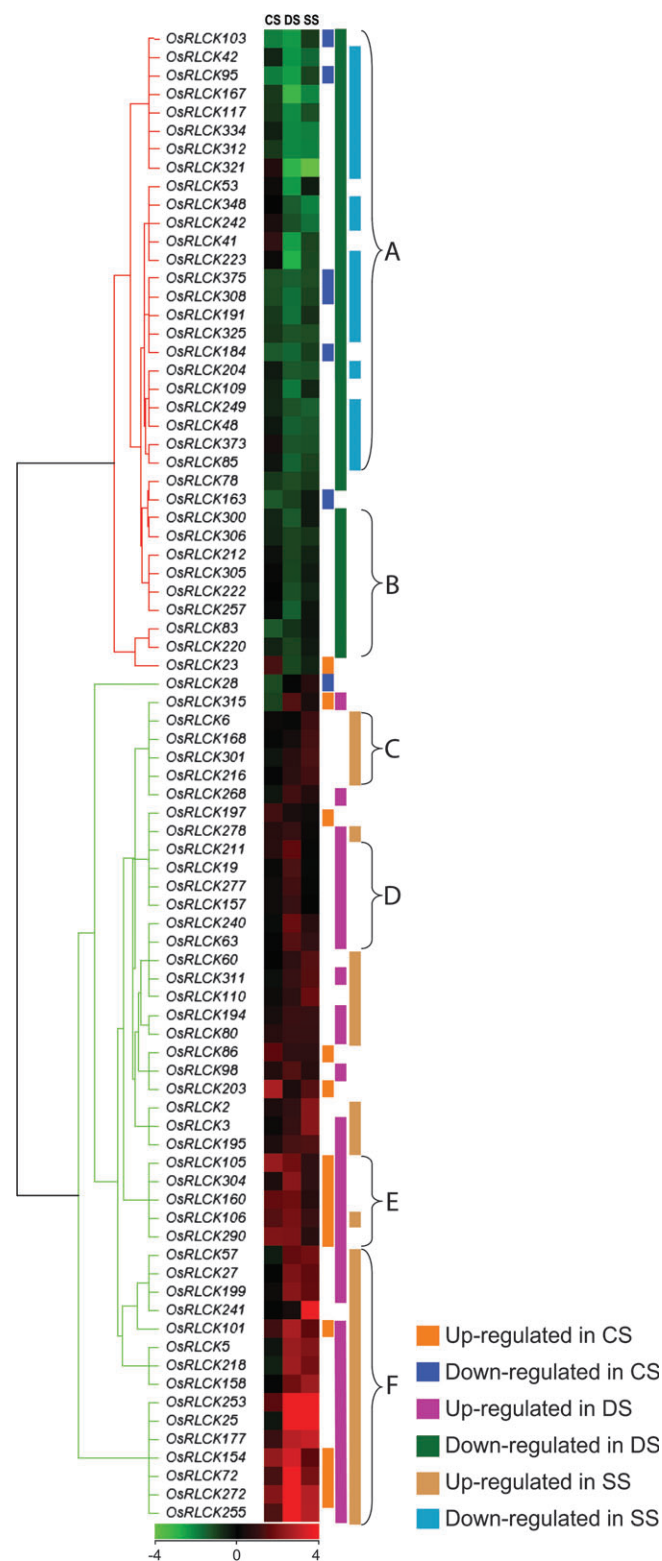


Figure 6. Expression Profiles of *OsRLCK*s under Three Abiotic Stress Conditions (CS: Cold Stress, DS: Dehydration Stress, SS: Salt Stress).

A gene is considered differentially expressed under abiotic stress conditions if it is up- or down-regulated at least two-fold, at P -value < 0.05 , with respect to 7-day-old seedling. Clustering

Conery, 2000; Prince and Pickett, 2002; He and Zhang, 2005; Cusack and Wolfe, 2007). Furthermore, segmentally duplicated *OsRLCK*s displayed a greater degree of functional divergence than tandemly duplicated genes, since neofunctionalization was more prevalent in segmentally duplicated genes and retention of expression was altogether absent. It is possible that similar expression of certain duplicated genes is due to local control regions (Li et al., 2007). Eukaryotic genes with multiple introns and exons are also regulated at the level of gene splicing. Alternative splicing, many times, leads to generation of new protein isoforms and thus increases the genome complexity (Smith and Valcarcel, 2000). Plants have been shown to display a greater degree of variety in alternative splicing that is mainly of the intron retention type (in animals, exon skip type is preferred) (Ner-Gaon et al., 2007). In rice, ~21.2% of the coding genome shows alternative splicing (Wang and Brendel, 2006) and its regulation by environmental stresses has been shown in another model plant, *Arabidopsis* (Tanabe et al., 2007). In our analysis, of 379 *OsRLCK*s, 56 genes (14.7% of the gene family) possibly undergo alternative splicing. It is interesting to note that alternative splicing may have far-reaching functional implications, as five genes (*OsRLCK28*, 185, 253, 306, and 375) were found to have alternatively spliced gene models with missing kinase domains encoding regions (Supplemental Figure 4). This may lead to further functional diversification of this large gene family.

Though *OsRLCK*s represent a highly relevant and considerably large rice gene family, none of its members has been functionally characterized. However, *RLCK*s from tomato (*Pto*) and *Arabidopsis* (*CDG1*, *CRCK1*, and *PBS1*) have been implicated in stress response and development (Tang et al., 1996; Swiderski and Innes, 2001; Muto et al., 2004; Yang et al., 2004). Their closest homologs in rice turned out to be *OsRLCK48*, 100 for *CRCK1*; *OsRLCK55*, 72 for *CDG1*; and *OsRLCK72*, 185 for *PBS1* on the basis of protein-level homology. Tomato *Pto* gene did not show any *RLCK* homolog in rice. *Arabidopsis CRCK1* is up-regulated by salt and cold stresses, while its homolog in rice, *OsRLCK48*, was down-regulated during these stresses in our experiment, though *OsRLCK100* showed marginal up-regulation in these

of expression profile was done with log transformed average values taking seedling as base-line. The color scale is given in Log₂ intensity value. The vertical color bars at the right side of the heat map represent genes differentially regulated more than two-fold in any one of the stress treatments with reference to seedling. Based on the expression pattern, genes were grouped into six categories: A, preferentially down-regulated in desiccation and/or salt stress; B, down-regulated in desiccation stress; C, up-regulated in salt stresses; D, up-regulated in desiccation stress; E, preferentially up-regulated in cold and/or desiccation stress; F, preferentially up-regulated in desiccation and/or salt stress.

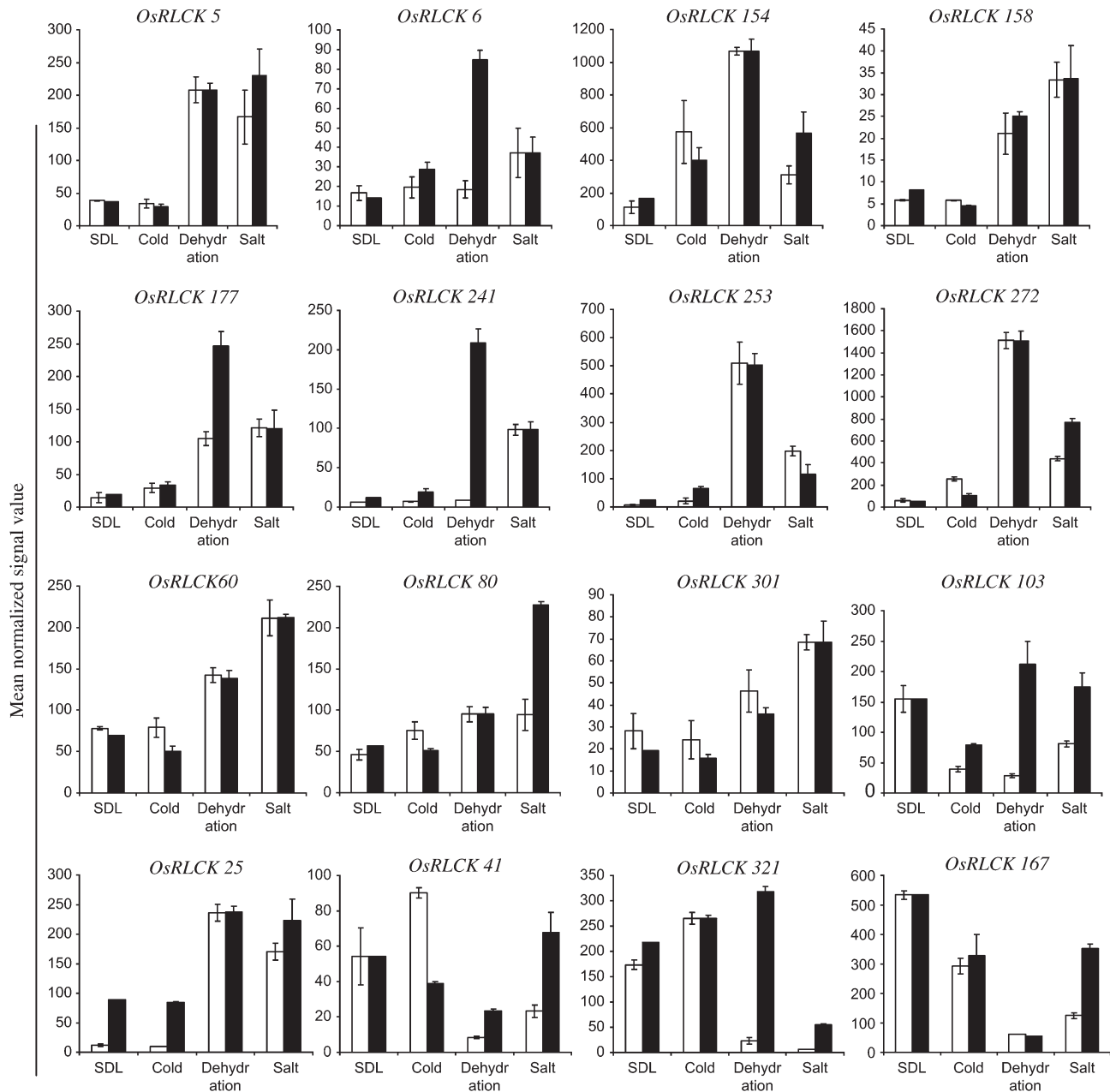


Figure 7. Validation of Selected *OsRLCKs* Expression Profiles by Q-PCR.

For real-time PCR analysis, the expression level of each gene was calculated in relation to its expression in unstressed 7-day-old seedlings and these values were normalized with the maximum average values obtained in the case of microarrays. White bars represent the expression from microarrays, while black bars represent the real-time PCR values. Error bar is shown. SDL, unstressed 7-day-old seedlings.

conditions. CDG1 is involved in brassinosteroid-mediated regulation of growth; its closet rice homolog, *OsRLCK72*, showed marginal differential expression during vegetative and reproductive development but was highly up-regulated during all three abiotic stresses. However, all these rice RLCK homologs showed up-regulation during biotic stresses. In our analysis, a significant proportion of the *OsRLCK* gene family showed differential expression in selected stages of

panicle and seed development. Interestingly, the majority of the genes also showed stage-specific expression. This temporal and spatial display of gene expression reflects the attainment of specialized functions by *OsRLCKs*.

We have shown that a higher number of *OsRLCKs* respond to biotic stress than abiotic stress, albeit with some common component. These observations reveal both general and specialized roles for *OsRLCKs* during

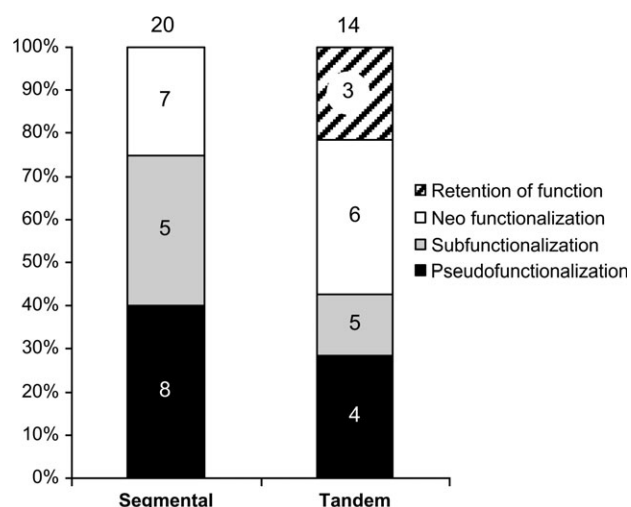


Figure 8. Expression Categories of *OsRLCKs* Present in Segmentally and Tandemly Duplicated Regions of Rice Chromosomes.

Based on microarray expression evidence, duplicated pairs were grouped into four different categories. Numbers in the stacked columns represent the number of duplicated pairs in the respective categories. For details of expression profiles, refer to Supplemental Figure 3.

stress. All abiotic stress-related *OsRLCKs* fall in two classes of up- and down-regulated genes. However, *OsRLCK315* was found to be both up-regulated in dehydration and down-regulated in cold stress. Confirmation of microarray expression profile by real-time PCR showed similar expression patterns except in a few samples. Quantitative and, to some extent, qualitative variations are often seen between results of these two techniques (Dallas et al., 2005). About 69% abiotic stress-responsive *OsRLCKs* were also found to be associated with abiotic stress QTLs. Receptor kinase in animals and RLKs in plants transduce signals by reversible protein phosphorylation (Ventura et al., 1994). Kinase-associated protein phosphatase (KAPP, a type 2C phosphatase) interacts with several RLKs in plants and regulates signaling by dephosphorylation (van der Knaap et al., 1999; Shiu and Bleeker, 2001b). In our microarray analysis, two rice orthologs of KAPP (LOC_Os7g11010 and LOC_Os03g59530) were found to be expressing differentially under abiotic stresses (data not shown), suggesting that KAPP may serve as a common regulator of RLK signaling.

It is interesting to note that, during reproduction and abiotic stress, ~31% *OsRLCKs* showed overlap of expression profile. Of the total *RLCKs* characterized for function in plants so far, only *Arabidopsis CRCK1* and wheatgrass *Esi47* have been shown to play a role in abiotic stresses tolerance (Shen et al., 2001; Yang et al., 2004). In this regard, our data provide new insights into the possible role of *OsRLCKs* during abiotic stress signaling in rice. Given the fact that the majority of *RLKs* are involved in biotic stress

response as reflected by greater expansion in this clade (Morillo and Tax, 2006) and expression of ~31% of rice *RLCKs* are affected by one or other type of biotic stress, the possibility of retention of functional similarities between these two groups of proteins is high. Furthermore, members of one rice clade (Clade 2) closely associated with yeast kinases were mainly involved in biotic stress response. This indicates that that clade may have originated from ancient kinases, while other clades could have result from subsequent lineage-specific expansion during evolution.

In conclusion, the present study provides insights into the phylogenetic relationship, organization, conservation of structural domains and their arrangements as also expression profiles of the *OsRLCK* gene family. Expression profiling of the *OsRLCK* gene family has unraveled their probable function during stress and development, which can be extended to genetic, physiological, and molecular analysis for elucidation of the specific function of target *RLCKs* in rice.

METHODS

Identification of *RLCKs* in the Rice Genome

Out of a total of 189 and 149 *RLCKs* reported from *indica* rice and *Arabidopsis*, respectively (Shiu and Bleeker, 2001a; Shiu et al., 2004), two of the rice *RLCKs* were found to be redundant and two accessions reported for *Arabidopsis* were no longer valid in the TAIR database (www.arabidopsis.org/), bringing the total number of rice and *Arabidopsis* *RLCKs* to 187 and 147, respectively. An HMM profile generated from the *RLCKs* (<http://hmmer.janelia.org/>; Eddy 1998) was used to scan version 5 of TIGR rice pseudo molecules data (Chromosomes 1–12) (<http://rice.plantbiology.msu.edu/pseudomolecules/info.shtml>) to search for such sequences. A total of 1866 unique sequences were identified using this approach. In an alternative approach, one representative from each of the previously identified RLK subfamilies in rice and *Arabidopsis* along with the genes in *Homo sapiens*, *Mus musculus*, *Rattus rattus*, *Bos Taurus*, *Danio rerio*, *Drosophila melanogaster*, *Gallus gallus*, *Pan troglodytes*, *Apis mellifera* as well as completely sequenced microbial genomes showing significant similarity to CLAVATA kinase domain were used to build an HMM profile, which was used to identify kinase domain containing proteins encoded by rice genome. This exercise added 16 more genes to the list of 1866 that had been identified by using rice and *Arabidopsis* specific HMM profiles. Domain analysis on these 1882 sequences using the SMART database could identify 1425 with at least one kinase domain. The remaining 457 sequences included 345 genes with TE signatures and thus discarded from the analysis. At this stage, we were left with 112 sequences that were identified by an HMM profile search but did not contain kinase domain according to SMART. An InterPro scan (www.ebi.ac.uk/InterProScan/) of these sequences identified 56 additional

sequences with at least one putative kinase domain. To further refine our search for non-TM kinases, sequences were analyzed in TMMH2 for prediction of transmembrane domains specifically. The 1481 kinase domain sequences along with 15 additional kinase domains (belonging to sequences with multiple kinase domains) and kinase domains from 52 kinase representatives (Shiu et al., 2001, 2004) were aligned using the ClustalX multiple alignment program (Thompson et al., 1997) and their relationship was viewed using NJ plot (APH (3') III used as an outgroup) in order to identify the clade(s) shared by RLCKs and RLKs. Cytoplasmic (lacking transmembrane domain) kinases, falling in the superclade specific to typical RLKs, were grouped as RLCKs. A similar analysis was done to search for cDNAs representing any additional RLCKs in the KOME database (<http://cdna01.dna.affrc.go.jp/cDNA/>). A total of 379 RLCKs identified in this way were aligned and their relationship viewed as described above. The phylogenetic relationships of the 379 *OsRLCKs* were studied by aligning the derived amino acid sequences by the ClustalX multiple alignment program (Thompson et al., 1997) and generating a radial tree using NJ plot and tree view software. Clades having $\geq 50\%$ bootstrap values were marked. The kinase domains of 379 *OsRLCKs* were aligned with those of yeast kinases (www.yeastgenome.org/) to make the phylogenetic comparison. The length of ORFs, number of introns, chromosomal location, expression evidence (cDNA or EST), and annotation of each of the short-listed genes as obtained from TIGR (<http://rice.plantbiology.msu.edu/index.shtml>) are summarized in Table 1. The TIGR segmental duplication database (http://rice.plantbiology.msu.edu/segmental_dup/100kb/segdup_100kb.shtml) was consulted to access the role of genome duplication events in the evolution of this class of genes as described previously (Agarwal et al., 2007). To calculate the sequence homology between tandemly duplicated *OsRLCKs*, the first gene of a group was compared with the other genes of that group, and, in the cases in which more than two members formed a group, by using MegAlign software 4.03© (DNASTAR).

Association of *OsRLCKs* with Abiotic Stress-Related QTLs

In order to study the association of the RLCK family with abiotic stress, the Gramene (www.gramene.org/) database was searched for QTLs in the abiotic stress category. Positions of all the 61 traits available under this category were mapped to the 12 rice chromosomes to identify the RLCKs associated with these QTLs. *RLCKs* found to be differentially expressed during abiotic stress and falling within the range of QTLs were identified.

Comparison of *OsRLCK* Family with *indica* Rice RLCKs

In order to establish a relationship between the previously reported *indica* rice RLCKs (Shiu et al., 2004) with the *japonica* rice RLCKs (our data), *OsRLCKs* (379) were used for BLAT

search against *indica* (187) RLCKs setting criteria of $\geq 90\%$ identity and $\geq 80\%$ coverage. These *indica* RLCKs had been identified from rice genome published by Yu et al. (2002). Since then, an improved *indica* rice genome has been published by Yu et al. (2005). This was also used to search for RLCKs using the same HMM models used for searching the *japonica* genome. The 1684 unique *indica* sequences identified using the HMM search were used for BLAT search with the *japonica* rice RLCKs using the same criteria as described above.

Microarray-Based Gene Expression Analysis

To generate the expression profile of *OsRLCKs*, whole genome microarray analysis was performed (Agarwal et al., 2007). The samples for the microarray experiment included three vegetative stages (mature leaf, root, and seedling), 11 reproductive stages (P1–P6 and S1–S5; stages during panicle and seed development, respectively, as described by Itoh et al., 2005), and three abiotic stress conditions (cold, salt, and dehydration). RNA was isolated from three biological replicates and microarray experiments were carried out using 51 Affymetrix GeneChip Rice Genome Arrays (Gene Expression Omnibus, GEO, platform accession number GPL2025) as described. The raw data (*.cel) files generated from all the chips were imported into Array Assist 5.0 software for detailed analysis (Stratagene, USA). To stabilize the variation of data from all the chips, normalization of the raw data was performed using GC-RMA, GeneChip Robust Multi-array Analysis, algorithm (Wu et al., 2003). Normalized signal intensity values were log transformed and averages of three biological replicates for each sample were used for analysis. Student's *t*-test was performed to identify differentially expressed genes (fold change ≥ 2 , at *P*-value < 0.05) with respect to all vegetative stages in the case of reproductive development and seedling in the case of stress samples. As only 1.5% of genes were found to be differentially expressed between replicates in all the stages examined, two-fold was set as the cut-off for differential expression. The data for only one probe set per gene (usually the 3' probeset) was used for the analysis. The data were base-line transformed by taking mature leaf and seedling as the base-lines for reproductive stages and stress samples, respectively. On the basis of expression profiles, genes were grouped by using self-organizing maps (SOM) and distance matrix Euclidian on rows (developmental expression) and both rows and columns (stress expression) with 100 maximum iterations.

MPSS Data Analysis

To analyze the expression patterns of *RLCKs* during biotic stress, Massively Parallel Signature Sequencing (MPSS) data (<http://mpss.udel.edu/rice/>) were retrieved for 17 base

signatures from selected libraries. Only those signatures that were unique to the genome, preferably in 3' UTR and transcribed from the respective strand of the gene, were included in the analysis (Classes 1 and 2). A TPM cut-off of ≥ 3 was set to avoid the background signal. The normalized transcript abundance values per million (TPM) were used to assess the expression profile.

Real-Time PCR Analysis

Real-time PCR reactions were carried out as described previously (Agarwal et al., 2007). Briefly, 2 μ g of total RNA from biological triplicates was used for cDNA synthesis using the High-Capacity cDNA Archive Kit (Applied Biosystems, USA) in 20 μ l reaction. 1 μ l of this cDNA was used for determining the expression level of a gene using SYBR Green PCR master mix (Applied Biosystems, USA) in the ABI Prism 7000 sequence detection system (Applied Biosystems, USA). Since *RLCKs* are members of a large gene family, each primer pair was checked in the TIGR and NCBI databases by blasting with rice pseudomolecules (Release 5.0) to ensure the uniqueness and specificity of amplification. Variations due to quality and number of RNA or cDNA samples were normalized using the *ACTIN1* gene as control. Relative mRNA levels of the genes were calculated as $\Delta\Delta C_T$ values in comparison to unstressed seedlings.

SUPPLEMENTARY DATA

Supplementary Data are available at *Molecular Plant Online*.

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